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Complete solutions for two-body collapse model of collisionless dark matter

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ABSTRACT

The merger of two bodies is the elementary step of the inverse mass cascade in dark matter flow. We solve the two-body collapse model (TBCM), two masses interacting through a power-law potential $V \propto R^n$ ($n = -1$ for gravity) in an expanding background, exactly. The problem reduces to one equation for an amplitude function $F_a(x)$ that measures the physical size of the pair, with three dimensionless parameters: n, a virial parameter γ and the ratio β of the Hubble velocity to the tangential velocity. Three types of solution exist: short-range forces ($n \leqslant -2$) let a pair expand with the background (type I) or fall to the centre (type III); long-range forces ($-2 < n < 0$) produce bound pairs that oscillate about a fixed physical size (type II), i.e. stable clustering, for every such n. We give the exact late-time (Kepler) limit, the energy cycle of a bound pair, the turnaround time, a perturbative solution, the exact free-fall time in the matter and radiation eras with its asymptotes, and phase diagrams of the three types. The critical overdensity $\Delta_c = 18\pi^2$ ($32\pi^2$ in the radiation era) is equivalent to an orbital period equal to the cosmic time. Mapping 956,476 two-particle haloes of an N-body simulation onto the model, 82 per cent are bound at $z = 0$; their mean radial velocity vanishes, as stable clustering requires, and their orbital phases are close to uniformly mixed where the orbits are resolved by the simulation. At $z \geqslant 2$ the pairs are still falling in. A generalized stable clustering hypothesis for odd and even moments of the pairwise velocity holds in the simulation for the third moment but not for higher ones, which shows that infall and random pairwise velocities are correlated.

Key words: dark matter – large-scale structure of Universe – methods: analytical – methods: numerical

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1 INTRODUCTION

Collisionless systems often show properties that point to common physical principles. The self-gravitating collisionless fluid dynamics (SG-CFD) studies the motion of collisionless matter under its own gravity; the large-scale collapse of dark matter is its most important example and the basis of the standard models of structure formation. Structure forms hierarchically, small structures merging into larger ones. The same process can be described by a halo-mediated inverse mass cascade, in which haloes pass their mass onto larger and larger haloes [1, 2]. Haloes form because a collisionless system with long-range interactions maximizes its entropy in this way [3, 4], and the cascade leads to the halo mass functions and density profiles of a stochastic random-walk theory [5, 6].

Over a finite time interval a merger may involve several bodies; over an infinitesimal interval dt it involves two and only two [7]. The collapse of two bodies in an expanding background is therefore the elementary step of the cascade. The two-body problem in a static background is the Kepler problem; in an expanding background it has no closed-form solution, and its full solution is the subject of this paper.

Two analytical tools dominate the description of nonlinear collapse. The spherical collapse model (SCM) follows a spherical overdensity [8, 9]; it was extended to self-similar collapse [10, 11], to non-radial orbits through a centrifugal term [12, 13] and to ellipsoidal collapse [14, 15], and it provides the critical overdensity $18\pi^2$ used by halo models [16, 17]. The stable clustering hypothesis (SCH) assumes that on small scales bound structures keep a fixed physical size, so that the mean pairwise peculiar velocity cancels the Hubble flow, $\langle \Delta u_L \rangle = -Hr$ [18, 19, 20]. Combined with the pair conservation equation it predicts the small-scale correlation function; it has been tested with simulations of limited resolution [21, 22, 23, 24].

The two-body collapse model (TBCM) connects these tools. It contains the nonradial motion that the SCM lacks and gives a dynamical picture of stable clustering. A first version [25] solved the

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model in a transformed system with a static background and constant damping. That transformation is exact only for gravity (Section 3.5). Here we solve the exact problem for every potential exponent n and every power-law cosmology, and we test it against an N-body simulation: the two-particle haloes of the simulation are mapped onto the solutions of the model, and the pairwise velocity moments test a generalized stable clustering hypothesis.

Section 2 recalls the cosmic energy equation and motivates a general exponent n. Section 3 formulates the TBCM and reduces it to one equation; Sections 4–6 treat the three types of solution; Section 7 maps the two-particle haloes of the simulation onto the model; Section 8 discusses stable clustering and its generalization, violent relaxation and the SCM; Section 9 concludes.

2 COSMIC ENERGY EVOLUTION

A system of N collisionless particles of mass m_p in an expanding background obeys, in comoving coordinates $\mathbf{x}$ and physical time t [20, p. 44],

$$\ddot{\mathbf{x}}_i + 2H\dot{\mathbf{x}}_i = -\frac{Gm_p}{a^3}\sum_{j\neq i}\frac{\mathbf{x}_i-\mathbf{x}_j}{|\mathbf{x}_i-\mathbf{x}_j|^3}, \tag{1}$$

with scale factor a and Hubble parameter $H=\dot{a}/a$. Its energy satisfies the cosmic energy (Layzer–Irvine) equation [26, 27]

$$\frac{\partial E_y}{\partial t} + H\left(2K_p + P_y\right) = 0, \tag{2}$$

where K_p is the specific peculiar kinetic energy, P_y the specific potential energy in physical coordinates and $E_y = K_p + P_y$. Equation (2) describes how energy is exchanged with the expanding background.

Gravity is the only interaction, yet the dark matter flow never reaches the virial equilibrium $2K_p = -P_y$ of an isolated system. To quantify the departure, consider a general power-law pair potential and force between particles at physical distance $R = ar$,

$$V_p = -\frac{G_n m_p^2}{R^{-n}}, \qquad F_p = -\frac{dV_p}{dR} = \frac{nG_n m_p^2}{R^{1-n}}, \tag{3}$$

with a generalized force constant G_n ($G_n = G$ and $n=-1$ for gravity). Long-range forces have $-2<n<0$, short-range forces $n\leqslant -2$. The equation of motion becomes

$$\ddot{\mathbf{x}}_i + 2H\dot{\mathbf{x}}_i = \frac{nG_n m_p}{a^{2-n}}\sum_{j\neq i}\frac{\mathbf{x}_i-\mathbf{x}_j}{|\mathbf{x}_i-\mathbf{x}_j|^{2-n}}, \tag{4}$$

which reduces to Eq. (1) for $n=-1$; the factor a^{2-n} (not a^3) follows from $R=ar$. The cosmic energy equation for Eq. (4) reads

$$\frac{\partial E_y}{\partial t} + H\left(2K_p - nP_y\right) = 0. \tag{5}$$

For a constant Ht and energies growing as $K_p, P_y \propto t^x$, Eq. (5) gives $2K_p = n_e P_y$ with an effective exponent

$$n_e = n - \frac{(n+2)x}{x+2Ht}. \tag{6}$$

In the matter era ($Ht=2/3$) with $x=1$, $n_e=(4n-6)/7$, i.e. $n_e=-10/7$ for gravity, and

$$K_p = -\varepsilon_u t, \qquad P_y = -\frac{2}{n_e}\varepsilon_u t = \frac{7}{5}\varepsilon_u t, \qquad E_y = \frac{2}{5}\varepsilon_u t, \tag{7}$$

where $\varepsilon_u<0$ is the rate at which energy is cascaded [1, 28]. The total energy decreases with time, as if it were dissipated, although there is no viscosity: a statistically steady state is maintained in which the system releases energy and increases its entropy by the mass and energy cascade. With $K_p=\frac{3}{2}u^2$ and the one-dimensional velocity dispersion $u_0 = 354.6\,\mathrm{km/s}$ of all particles at $z=0$ of the simulation used below (SCDM, $h=0.5$, age $t_0=2/3H_0=13.0\,\mathrm{Gyr}$), $\varepsilon_u = -\frac{3}{2}u_0^2/t_0 = -4.6\times10^{-7}\,\mathrm{m^2\,s^{-3}}$. In the simulation u^2 grows as $t^{0.84}$ for $z\leqslant 2$ rather than t, which gives $n_e=-1.39$ for gravity from Eq. (6) and an energy transfer rate $\partial K_p/\partial t = 0.84K_p/t$, i.e. $\varepsilon_u \approx -3.9\times10^{-7}\,\mathrm{m^2\,s^{-3}}$ at present. Either way, the effective exponent of the self-gravitating flow differs from -1, which motivates studying the elementary two-body problem for a general n.

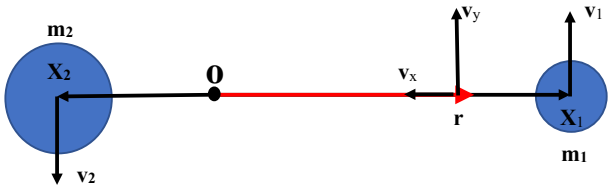


Figure 1. Two-body collapse in an expanding background. Two masses m_1 and m_2 are separated by $2\mathbf{r}$, where $\mathbf{r}$ is the comoving displacement vector; $\mathbf{v}_1$ and $\mathbf{v}_2$ are the peculiar velocities of the masses and v_x, v_y the components of the peculiar velocity of $\mathbf{r}$.

3 THE TWO-BODY COLLAPSE MODEL

3.1 Equations of motion

Two masses m_1 and m_2 with comoving positions $\mathbf{x}_1$ and $\mathbf{x}_2$ and a separation $2\mathbf{r}$, $\mathbf{r}=(\mathbf{x}_1-\mathbf{x}_2)/2$ (Fig. 1), obey (from Eq. 4)

$$\ddot{\mathbf{x}}_1 + 2H\dot{\mathbf{x}}_1 = \frac{nG_n m_2}{(2r)^{1-n}a^{2-n}}\hat{\mathbf{r}}, \qquad \ddot{\mathbf{x}}_2 + 2H\dot{\mathbf{x}}_2 = -\frac{nG_n m_1}{(2r)^{1-n}a^{2-n}}\hat{\mathbf{r}}. \tag{8}$$

The centre of mass $\mathbf{X}=(m_1\mathbf{x}_1+m_2\mathbf{x}_2)/M$, $M=m_1+m_2$, obeys $\ddot{\mathbf{X}}+2H\dot{\mathbf{X}}=0$ and is placed at the origin, so that $\mathbf{x}_1=\mu\mathbf{r}$ and $\mathbf{x}_2=-(2-\mu)\mathbf{r}$ with $\mu=2m_2/M$. The displacement vector obeys

$$\ddot{\mathbf{r}} + 2H\dot{\mathbf{r}} = \frac{nG_n M}{2(2r)^{1-n}a^{2-n}}\hat{\mathbf{r}}, \tag{9}$$

which has the form of a damped oscillator, $\ddot{\mathbf{r}}+(c/m)\dot{\mathbf{r}}=-(k/m)\mathbf{r}$, with a time-dependent damping $2H$ and a nonlinear restoring force. Multiplying by a gives the equivalent form for the physical half-separation $\mathbf{R}=a\mathbf{r}$,

$$\ddot{\mathbf{R}} = \frac{\ddot{a}}{a}\mathbf{R} + \frac{nG_n M}{2(2R)^{1-n}}\hat{\mathbf{R}}, \tag{10}$$

in which the first term is the force of the homogeneous background.

The motion is planar. Writing $\mathbf{r}=r(t)\,e^{i\omega(t)t}$ with a time-dependent frequency $\omega(t)$, Eq. (9) separates into

$$\ddot{r} + \underbrace{2H\dot{r}}_{1} - \underbrace{\frac{nG_n M}{2(2r)^{1-n}a^{2-n}}}_{2} = \underbrace{r(\omega+t\dot{\omega})^2}_{3}, \qquad \frac{\dot{r}}{r} = -H - \frac{1}{2}\frac{\partial\ln(\omega+t\dot{\omega})}{\partial t}, \tag{11}$$

with the damping (term 1), the interaction (term 2) and the centrifugal force (term 3). The initial conditions at the time t_i when the collapse starts (scale factor a_i, Hubble parameter H_i) are

$$r(t_i)=r_i, \qquad \dot{r}(t_i)=v_x/a_i, \qquad \omega(t_i)=0, \qquad \dot{\omega}(t_i)=v_y/(r_i t_i a_i),$$

(12)

where v_x and v_y are the radial and tangential peculiar velocities of $\mathbf{r}$. We introduce the velocity scale of a circular orbit at t_i,

$$\tilde{v}^2 = \frac{-nG_nM}{4(2a_ir_i)^{-n}}, \tag{13}$$

and a frequency function

$$F(t) = (\omega + t\dot{\omega})^{-1/2} = \left(\frac{\partial(\omega t)}{\partial t}\right)^{-1/2}. \tag{14}$$

The second of Eqs. (11) then integrates to

$$r(t) = \left(v_y r_i a_i\right)^{1/2} F(t)\, a^{-1}, \tag{15}$$

so that F measures the physical size ar of the pair, and the angular momentum $a^2r^2\dot{\theta} = v_y r_i a_i$ is conserved. With the identity $\ddot{r}/r = \partial_t(\dot{r}/r) + (\dot{r}/r)^2$, Eqs. (11) combine into one equation for F without a first derivative:

$$\begin{aligned} \ddot{F} &= \underbrace{F\left(\dot{H}+H^2\right)}_{1} - \underbrace{\gamma^{n+2}\left(\frac{H_i}{\beta}\right)^{1+n/2} F^{n-1}}_{2} + \underbrace{F^{-3}}_{3}, \\ F(t_i) &= \left(\frac{\beta}{H_i}\right)^{1/2}, \qquad \dot{F}(t_i) = \left(\frac{v_x}{v_y}+\beta\right)\left(\frac{H_i}{\beta}\right)^{1/2}, \end{aligned} \tag{16}$$

where the seven parameters of the problem (n, H, M, r_i, v_x, v_y, t_i) are grouped into

$$\gamma = \left(\frac{\tilde{v}^2}{v_y^2}\right)^{1/(n+2)}, \qquad \beta = \frac{H_i a_i r_i}{v_y}. \tag{17}$$

The pair starts in virial balance ($v_y = \tilde{v}$) for $\gamma = 1$; β measures the Hubble velocity against the tangential velocity. The first term of Eq. (16) is $\ddot{a}F/a$, the force of the background. Both parameters fix the density of the pair, $\rho_{TB} = M/[4\pi(2a_ir_i)^3/3]$, relative to the background, $3H_i^2/(8\pi G)$:

$$\Delta = \frac{2^{-n}\gamma^{n+2}}{\beta^2}\frac{G}{G_n(a_ir_i)^{n+1}}, \qquad \Delta = \frac{2\gamma}{\beta^2} \text{ for gravity}, \tag{18}$$

which for gravity is independent of the size of the pair.

3.2 Reduction to one equation

When the damping term is small, term 2 of Eq. (16) balances term 3 at the mean value $F_m = \gamma^{-1}\beta^{1/2}H_i^{-1/2}$, and Eq. (14) gives the mean frequency $\omega_m = F_m^{-2} = \gamma^2 H_i/\beta$. Writing $F(t) = F_m F_a(x)$ with $x = \omega_m t$ reduces Eq. (16) to an equation for the amplitude function F_a:

$$\frac{d^2F_a}{dx^2} = \underbrace{X\frac{F_a}{x^2}}_{1} - \underbrace{F_a^{n-1}}_{2} + \underbrace{F_a^{-3}}_{3}, \qquad X = \left(\dot{H}+H^2\right)t^2, \tag{19}$$

with the initial conditions

$$F_a(x_0) = \gamma, \qquad F_a'(x_0) = \frac{1}{\gamma}\left(\frac{v_x}{v_y}+\beta\right), \qquad x_0 = \omega_m t_i = \frac{\gamma^2 H_i t_i}{\beta}. \tag{20}$$

In a power-law cosmology, $a \propto t^{Ht}$ with $Ht = 2/3$ (matter era) or $1/2$ (radiation era), $\dot{H} = -H/t$ and $X = Ht(Ht-1)$ is a constant. Equation (19) is exact: it is the simplest representation of the two-body problem in an expanding background, with four dimensionless parameters (n, γ, β, v_x/v_y) besides the cosmology. Once F_a is known,

$$\begin{aligned} \omega(t)\,t &= \int_{x_0}^{x} F_a^{-2}\,dx, \qquad \frac{ar}{a_ir_i} = \frac{F_a}{\gamma}, \\ r(t) &= \frac{\tilde{v}\beta}{\gamma^{2+n/2}H_i}F_a(x)\,a^{-1}, \end{aligned} \tag{21}$$

so that F_a is the physical size of the pair in units of a_ir_i/γ. We solved Eq. (19) and Eq. (9) independently for $n = -0.5, -1, -1.5$ and several initial states: the sizes agree to 2×10^{-10}. Unless stated otherwise we take $v_x = 0$ (the pair starts with the Hubble flow) and the matter era.

3.3 Energy and momentum

The specific potential and peculiar kinetic energies of the pair follow from F_a,

$$\begin{aligned} P_y &= \mu_m\frac{\tilde{v}^2}{n}\gamma^{-n}F_a^n, \\ K_p &= \mu_m\frac{\tilde{v}^2}{2}\gamma^{-n}\left[F_a^{-2} + \left(F_a' - \frac{Ht}{x}F_a\right)^2\right] \\ &= \mu_m\frac{\tilde{v}^2}{2}\gamma^{-n}F_a^{-2}\left[1 + \left(\frac{G}{|\mathbf{H}|}\right)^2\right], \end{aligned} \tag{22}$$

with the mass factor $\mu_m = 4m_1m_2/M^2$. Here G is the specific radial momentum (virial quantity) and $\mathbf{H}$ the specific angular momentum,

$$\begin{aligned} G &= \frac{\sum m_i(a\mathbf{x}_i)\cdot\mathbf{v}_i}{\sum m_i} = \mu_m a^2 r\dot{r} = |\mathbf{H}|\,F_a\left(F_a' - \frac{Ht}{x}F_a\right), \\ \mathbf{H} &= \frac{\sum m_i(a\mathbf{x}_i)\times\mathbf{v}_i}{\sum m_i} = \mu_m v_y a_i r_i\,\hat{\mathbf{z}}, \end{aligned} \tag{23}$$

where $\mathbf{v}_i = a\dot{\mathbf{x}}_i$ are peculiar velocities. The angular momentum is conserved. The term $(Ht/x)F_a$ is the Hubble flow in the units of F_a', so F_a' is the physical radial velocity and $F_a' - (Ht/x)F_a$ the peculiar one. The energies of Eq. (22) agree with a direct integration to 2×10^{-9}, and they satisfy the cosmic energy equation of the pair, Eq. (5), to a relative residual of 3×10^{-7}.

3.4 Three types of solutions

Figures 2 and 3 show solutions of Eq. (19) for pairs that start in virial balance ($\gamma = 1$, $\beta = 1$) with different exponents. Three types of solution exist:

(i) Type I: F_a grows monotonically, $F_a \propto x^{Ht} \propto a$ at large x. The pair expands with the universe at a fixed comoving size. This is the solution of short-range forces ($n \leqslant -2$) when the expansion dominates.

(ii) Type II: F_a grows in a linear regime, $F_a \propto x^{Ht}$, turns around and oscillates about a fixed value, i.e. the pair oscillates about a fixed physical size. This is the solution of long-range forces ($-2 < n < 0$) and the dynamical basis of stable clustering.

(iii) Type III: F_a vanishes at a finite time: the two bodies fall to the centre. This requires $n < -2$ when the angular momentum is finite (for $n = -2$ the interaction and the centrifugal force cancel exactly), or zero angular momentum ($v_y = 0$) for any n.

Figure 4 maps the types in the (β, γ) plane. For short-range forces the boundary is $\gamma \approx 1$ at small β and follows $\gamma \propto \beta^{2/(n+2)}$ at large β (Section 6); no oscillating solution exists. For long-range forces,

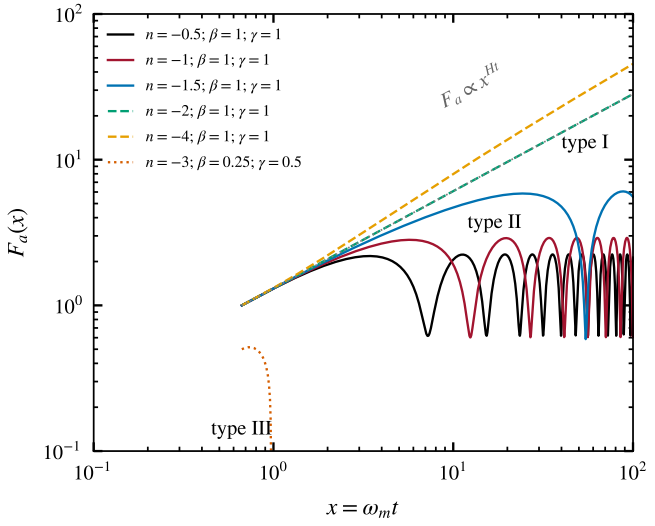


Figure 2. Amplitude function $F_a(x)$ (the physical size of the pair, Eq. 21) for different force exponents n, $Ht = 2/3$, $v_x = 0$. Type I (dashed): short-range forces, $F_a \propto x^{Ht} \propto a$, a fixed comoving size. Type II (solid): long-range forces, a linear regime $F_a \propto x^{Ht}$ followed by oscillations about a fixed physical size. Type III (dotted): a fall to the centre.

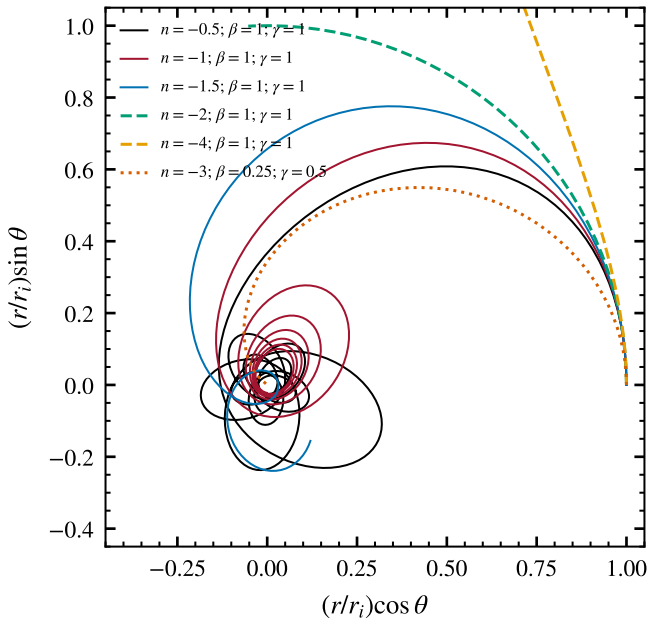


Figure 3. Trajectories of the comoving displacement vector $\mathbf{r}$ (Eq. 21) for the cases of Fig. 2, in units of r_i.

pairs with small γ (a tangential velocity well above the circular one) keep expanding for a long time; all others turn around and become bound. On a grid of 41×41 points ($10^{-2} \leqslant \beta, \gamma \leqslant 10^2$) our classification agrees with the author's independent MATLAB maps at 100 and 100 per cent of the points for $n = -3$ and -2.5, and at 94.6 and 87.3 per cent for $n = -1$ and -0.5. For $n = -1$, 79 of the 91 differing points are pairs that turn around later than $10^3 t_i$ (median $5 \times 10^5 t_i$); the other 12, at $\gamma \geqslant 20$, turn around at $t/t_i = 1.003$–923 in two independent integrations, while the author's map labels them

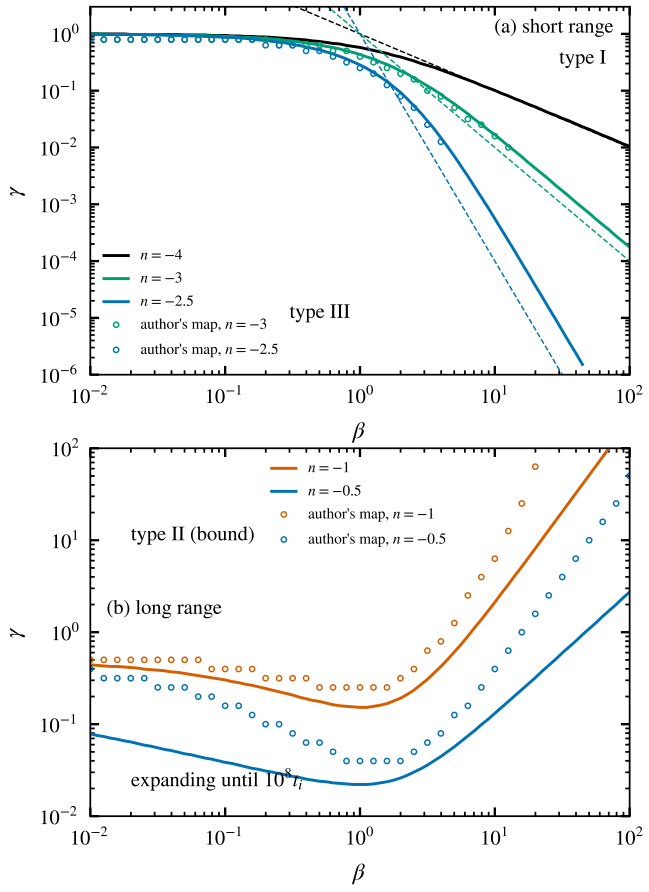


Figure 4. Phase diagrams in the (β, γ) plane, $Ht = 2/3$, $v_x = 0$. (a) Short-range forces: boundary between type I (above) and type III (below), with the asymptote $\gamma = \beta^{2/(n+2)}$ (dashed, $\lambda = 1$ in Eq. 41). (b) Long-range forces: lowest γ for which the pair turns around before $t = 10^8 t_i$. Circles: boundaries of the author's independent MATLAB maps.

as expanding. For $n = -0.5$ all 214 differing points turn around later than $1 \times 10^3 t_i$.

3.5 A transformed time

A useful time variable [25] is s with $ds/dt = a^{-3/2}$ (matter era), in which Eq. (9) takes the form

$$\frac{d^2\mathbf{r}}{ds^2} + \frac{H_0}{2}\frac{d\mathbf{r}}{ds} = \frac{nG_nMa^{n+1}}{2(2r)^{1-n}}\hat{\mathbf{r}} \tag{24}$$

with a constant damping $H_0/2$ ($H^2a^3 = H_0^2$). The factor a^{n+1} is essential: without it the transformed system is exact only for gravity. For $n = -0.5$ and -1.5 the system without the factor differs from the exact problem by 3.4 and 1.8 (maximum relative difference of the comoving size between t_i and $20t_i$, $\gamma = 1$, $\beta = 0.5$), while for $n = -1$ the difference is 1×10^{-9}; with the factor restored the agreement is 7×10^{-10}. Two consequences of dropping the factor, that stable clustering would exist only for gravity and that a critical β would exist only for discrete exponents, are therefore not properties of the two-body problem (Section 5). The transformed time remains useful for gravity and for the free fall (Section 6).

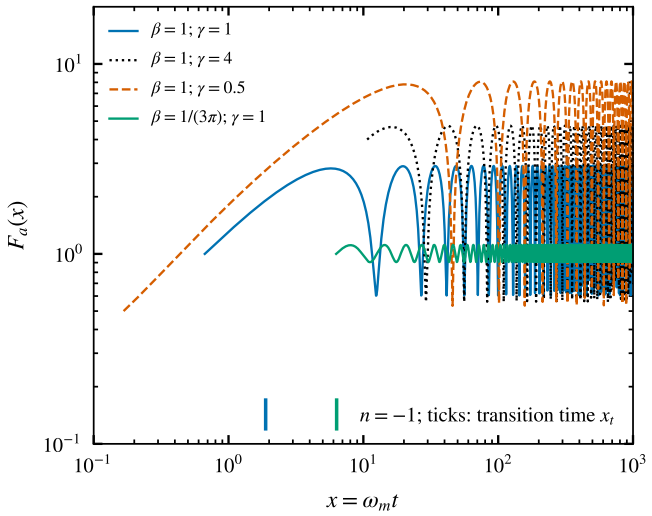


Figure 5. Type II solutions for gravity ($n=-1$), $Ht=2/3$, $v_x=0$, for several β and γ. Vertical ticks: transition time x_t (Eq. 27) where it exists. For small γ the linear regime is long and the solution approaches type I; for large γ the pair turns around at once.

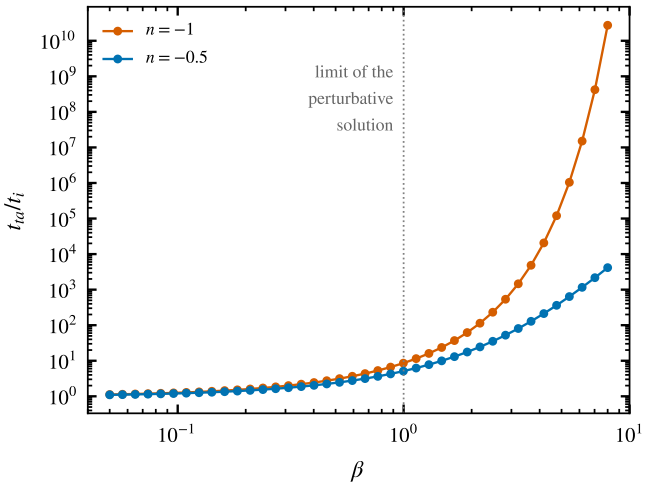


Figure 6. Turnaround time of a pair that starts with the Hubble flow in virial balance ($\gamma=1$, $v_x=0$), $Ht=2/3$. The dotted line marks $\beta=1$, the limit of validity of the perturbative solution (Section 5.5).

4 TYPE I SOLUTIONS: SHORT-RANGE FORCES

For $n=-2$ the interaction (term 2) and the centrifugal force (term 3) of Eq. (19) cancel, and the damping term alone determines the solution. With $v_x=0$ the exact solution is

$$F_a(x)=\gamma\left(\frac{x}{x_0}\right)^{Ht}\propto a, \tag{25}$$

which satisfies both initial conditions because $F_a'(x_0)=Ht\gamma/x_0=\beta/\gamma$. Since $X=Ht(Ht-1)<0$, the curvature of F_a is negative and tends to zero. For $n<-2$, terms 2 and 3 become negligible against term 1 at large x if the pair has not collapsed before, so that Eq. (25) holds asymptotically: the pair expands with the universe at a fixed comoving size (Eq. 21). Short-range forces have no oscillating solution. From Eqs. (22) and (23), the type I solution has

$$\begin{aligned} P_y&=\mu_m\frac{\tilde v^2}{n}\left(\frac{x}{x_0}\right)^{nHt}\propto a^n, \qquad G=0,\\ K_p&=\mu_m\frac{\tilde v^2}{2}\gamma^{-2-n}\left(\frac{x}{x_0}\right)^{-2Ht}\propto a^{-2}. \end{aligned} \tag{26}$$

A short-range force falls to the centre (type III) when it dominates the expansion, i.e. for a small γ and β: a small tangential velocity, a large mass or force constant, or a low Hubble parameter at the start. The boundary between types I and III (Fig. 4a) is $\gamma\approx1$ for small β and follows $\gamma=\beta^{2/(n+2)}$, i.e. $\lambda=1$ in Eq. (41), for large β; the slopes -1, -2 and -4 for $n=-4$, -3 and -2.5 are those of the boundaries in Fig. 4a.

5 TYPE II SOLUTIONS: LONG-RANGE FORCES

5.1 Linear regime and transition

At early times (small x) the damping term dominates and the solution follows the linear regime of Eq. (25), $F_a\approx\gamma(x/x_0)^{Ht}$: the pair expands with the background. The transition to the nonlinear regime occurs when terms 2 and 3 become comparable to term 1. Inserting the linear solution into $|XF_a/x^2|=|-F_a^{n-1}+F_a^{-3}|$ gives the transition time x_t,

$$\left|\left(\frac{x_t}{x_0}\right)^{2-4Ht}-\gamma^{n+2}\left(\frac{x_t}{x_0}\right)^{2+Ht(n-2)}\right|=\left(\frac{1}{Ht}-1\right)\beta^2. \tag{27}$$

For $n=-1$ and $\gamma=\beta=1$, $x_t=1.89$ (Fig. 5). After the transition the pair turns around and oscillates.

5.2 Turnaround

Every type II pair turns around eventually. For gravity and $\gamma=1$ the turnaround time grows steeply with β (Fig. 6): $t_{ta}/t_i=8.6$, 76 and 835 for $\beta=1$, 2 and 3, 8.2×10^6 for $\beta=6$ and 2.7×10^{10} for $\beta=8$; for $n=-0.5$, 5.1, 64 and 4157 for $\beta=1$, 3 and 8. The value $\beta=(2+n)^{3/2}$ (i.e. 1 for gravity) might be taken as the largest β for which an equilibrium collapse exists. It is, however, the limit of validity of the perturbative solution of the transformed system (Section 5.5); it is not a threshold of the exact problem, in which pairs with $\beta>1$ also become bound, only much later.

5.3 Late times: the static limit

Term 1 of Eq. (19) decays as x^{-2}. A bound pair therefore approaches the solution of the problem without expansion,

$$\frac{d^2F_a}{dx^2}=-F_a^{n-1}+F_a^{-3}, \tag{28}$$

which conserves $E=\frac12F_a'^2+\frac12F_a^{-2}+F_a^n/n$ and has bounded oscillating solutions for every $-2<n<0$. Because F_a is the physical size (Eq. 21), a bound pair oscillates about a fixed physical size for every long-range exponent: stable clustering is a property of all long-range forces, not only of gravity. For gravity with $F_a(x_0)=\gamma_s$ and $F_a'(x_0)=\beta_s/\gamma_s$, Eq. (28) is the Kepler problem and has the exact solution

$$\begin{aligned} x=x_0+\frac{1}{C_1^{3/2}}\Big[&\sqrt{C_1}\,\beta_s\mp\sqrt{C_1}\,Y(x)+\arctan\frac{1-C_1\gamma_s}{\sqrt{C_1}\,\beta_s}\\ &\mp\arctan\frac{1-C_1F_a(x)}{\sqrt{C_1}\,Y(x)}\Big], \end{aligned} \tag{29}$$

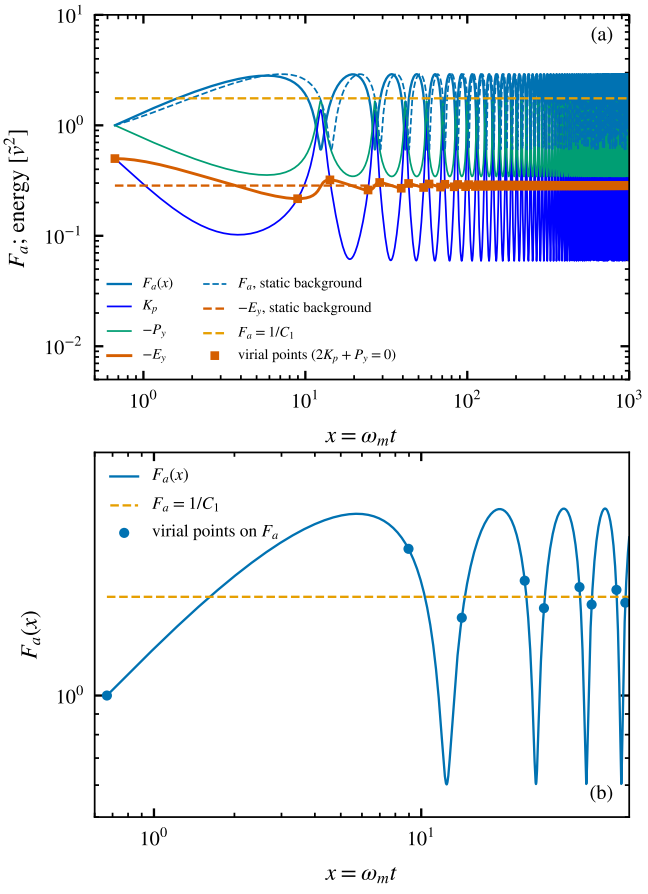


Figure 7. (a) Physical size F_a, kinetic energy K_p, potential energy $-P_y$ and total energy $-E_y$ of a pair with $n = -1$, $\gamma = \beta = 1$ (solid), in units of $\tilde{v}^2$ (mass factor omitted), and the static solution with the same late orbit (dashed, $\gamma_s = 1, \beta_s = 0.656$). Squares: virial points $2K_p + P_y = 0$, the extrema of E_y. (b) F_a with the virial points (circles) and the mean $1/C_1 = 1.756$ of the limiting Kepler orbit.

where the lower signs hold on the expanding branch ($F_a' > 0$) and the upper signs on the collapsing branch, and

$$C_1 = 1 - \frac{\beta_s^2 + (\gamma_s - 1)^2}{\gamma_s^2} \leqslant 1,$$
$$Y(x) = \sqrt{-C_1 F_a^2 + 2F_a - 1} = F_a |F_a'|. \tag{30}$$

The energy is $E = -C_1/2$, and a bound orbit requires $C_1 > 0$. F_a oscillates between

$$F_{a,\min/\max} = \frac{1 \mp \sqrt{1 - C_1}}{C_1}, \qquad T = \frac{2\pi}{C_1^{3/2}}, \tag{31}$$

about the mean $1/C_1$ (the semi-major axis), with period T in x and eccentricity $e = \sqrt{1 - C_1}$. For $n = -1$, $\gamma = \beta = 1$ the late oscillation of the expanding solution has $F_{a,\min} = 0.604$ and $F_{a,\max} = 2.908$, i.e. $C_1 = 0.570$, and it coincides with the static solution with $\gamma_s = 1$, $\beta_s = 0.656$ (Fig. 7).

5.4 Energy cycle

In the static background the energy of the pair is conserved. In the expanding background it is not: by Eq. (5) E_y decreases while $2K_p + P_y > 0$ and increases while $2K_p + P_y < 0$, so that its extrema are the virial points $2K_p + P_y = 0$ (Fig. 7). In the late Kepler orbit, $2K_p + P_y \propto 1/F_a - C_1$, so the collapse phase (energy decreasing) is the part of the orbit inside $F_a = 1/C_1$ around the pericentre and the expanding phase the part outside. With the eccentric anomaly of the orbit, their durations in x are

$$T_{1,2} = \frac{\pi \mp 2\sqrt{1 - C_1}}{C_1^{3/2}}, \tag{32}$$

and over each phase $|E_y|$ changes approximately as a power law, $|E_y| \propto x^{k_{1,2}}$, with

$$k_{1,2} = \pm \frac{4Ht\sqrt{1 - C_1}}{\pi \mp 2\sqrt{1 - C_1}}, \tag{33}$$

so that $k_1 T_1 + k_2 T_2 = 0$: the gain and loss over a period cancel and the energy approaches that of the static orbit. For the solution of Fig. 7, the phases last $T_1 = 4.26$ and $T_2 = 10.36$ and the exponents are $k_1 = 0.956$ and $k_2 = -0.393$, as predicted by Eqs. (32) and (33). The collapse phase resembles the decay of energy in turbulence, here driven by the expansion rather than by viscosity; in contrast to a dissipative system, the energy oscillates about its limiting value.

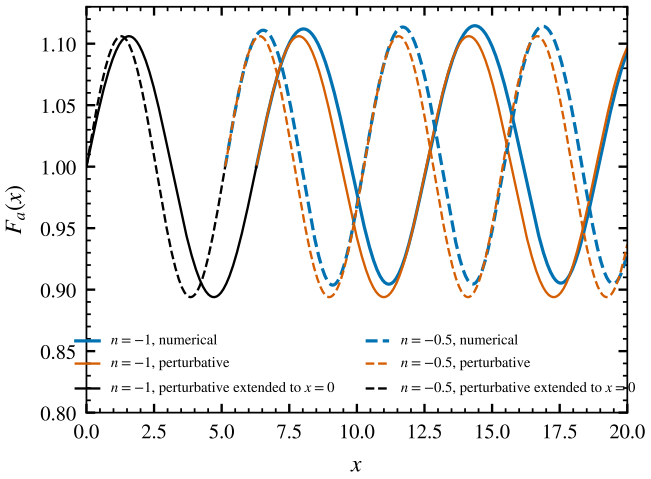


Figure 8. Numerical solutions (from x_0) and the perturbative solution Eq. (34) for $\gamma = 1$ and the critical $\beta = \beta_c$ (Eq. 35) for $n = -1$ and -0.5; black: the perturbative solution continued back to $x = 0$.

5.5 Perturbative solution and critical overdensity

For a pair in virial balance ($\gamma = 1$) with $\beta \ll 1$, neglecting term 1 of Eq. (19) and linearizing about $F_a = 1$ gives a harmonic solution that satisfies both initial conditions,

$$F_a(x) \approx 1 + \frac{\beta}{\sqrt{2+n}} \sin\left[\sqrt{2+n}\,(x - x_0)\right], \qquad x_0 = \frac{H_i t_i}{\beta}. \tag{34}$$

The amplitude is correct to first order in β (Fig. 8); the phase drifts slowly because term 1 is neglected (maximum error 0.017 over ten periods for $\beta = 0.05$). For a typical halo with a size of order 1 Mpc and a virial velocity of order 600 km s^{-1}, $\beta = Hr_h/\tilde{v} \approx 0.12$.

Consider haloes that grow continuously by mergers with an infinitesimal waiting time, so that a pair formed at t_i is in the same state as if it had formed at any earlier time; in the limit $t_i \to 0$ the perturbative solution continued back to $x = 0$ must start in the same

equilibrium, $F_a(0) = 1$ with the same phase as at x_0 (Fig. 8). This requires $\sqrt{2+n}\,x_0 = 2\pi$, i.e.

$$\beta_c = \frac{\sqrt{2+n}\,H_i t_i}{2\pi}, \tag{35}$$

which exists for every long-range exponent; $\beta_c = 0.1061 = 1/(3\pi)$ for gravity in the matter era and $0.0796 = 1/(4\pi)$ in the radiation era. Without the factor a^{n+1} in the transformed system (Section 3.5) a critical value would appear only for a discrete set of exponents, $n = (2-6m)/(1+3m)$, because the phase would then not grow linearly in time. With Eq. (18), the density of such pairs relative to the background is, for gravity,

$$\Delta_c = \frac{2}{\beta_c^2} = \frac{8\pi^2}{(Ht)^2} = \begin{cases} 18\pi^2 & Ht = 2/3, \\ 32\pi^2 & Ht = 1/2, \end{cases} \tag{36}$$

the critical overdensity of the SCM. Equation (36) is an identity: for $\gamma = 1$ the orbital frequency of the pair is $\omega_m = H_i/\beta$, and $\beta = \beta_c$ means that the orbital period $2\pi\beta/H_i$ equals the cosmic time t_i. The virialized sphere of the SCM, with density $18\pi^2\bar\rho$, has the same property: the period of a circular orbit at its edge, $2\pi(R^3/GM)^{1/2}$, equals t in the matter era, because $GM/R^3 = 4\pi G\rho/3 = 4\pi^2/t^2$ for $\rho = 18\pi^2\bar\rho$ and $\bar\rho = 1/(6\pi G t^2)$. Both models thus obtain $18\pi^2$ from the same condition.

At t_i a pair with $\beta = \beta_c$ moves with the Hubble velocity $H_i a_i r_i$ radially and with v_y tangentially, so that the angle between its velocity and its separation vector is $90° + \arctan\beta_c = 96.06°$ for gravity. For a merger with a large halo this is the angle of incidence of the merging mass at the halo surface [29]; it is relevant to the interpretation of a critical acceleration in dark matter haloes [30].

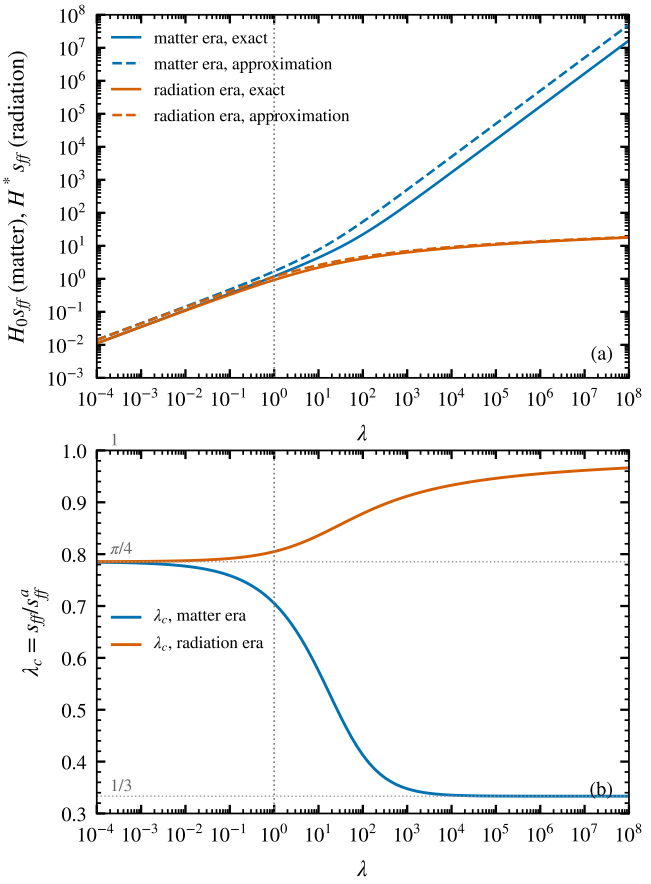


Figure 9. Free fall of a pair that starts at rest in comoving coordinates (Hubble flow), gravity. (a) Free-fall time in the transformed time, exact (solid) and the approximation of a constant force (dashed, Eqs. 40 and 44), for the matter and radiation eras. (b) Correction factor $\lambda_c = s_{ff}/s_{ff}^a$; dotted lines: the limits $\pi/4$, 1/3 and 1. The dotted vertical line marks $\lambda = 1$, the boundary between small and large scales.

6 TYPE III SOLUTIONS: FREE FALL

A pair falls to the centre when the interaction dominates both the expansion and the angular momentum: for short-range forces with small γ and β (Fig. 4a), and for any force when the tangential velocity vanishes. The free-fall time of a pair at rest in a static background follows from Kepler's laws,

$$t_{ff}^{ex} = \frac{\pi(a_i r_i)^{3/2}}{\sqrt{GM}} = \frac{\sqrt{2}}{8} T_{ri} = \sqrt{\frac{3\pi}{32 G\rho_{TB}}}, \qquad T_{ri} = \frac{2\pi a_i r_i}{\tilde v}, \tag{37}$$

where T_{ri} is the period of a circular orbit and ρ_{TB} the density of the pair. t_{ff}^{ex} depends only on the density. In an expanding background the free-fall time also depends on when the fall starts, because the Hubble flow resists the collapse; the effect is large on large scales and early in the history of the universe.

6.1 Transformed time

With a time s, $ds/dt = a^{-2y}$, Eq. (9) becomes

$$\frac{d^2\mathbf{r}}{ds^2} + 2H(1-y)a^{2y}\frac{d\mathbf{r}}{ds} = \frac{nG_nM}{2(2r)^{1-n}a^{2-n-4y}}\,\hat{\mathbf{r}}, \tag{38}$$

and the function $F^*(s) = F(t)\,a^{-y}$ obeys

$$\frac{d^2F^*}{ds^2} = F^* a^{4y}\left[(1-y)\dot H + (1-y^2)H^2\right] - \gamma^{n+2}\left(\frac{H_i}{\beta}\right)^{1+n/2} F^{*n-1}a^{y(n+2)} + F^{*-3}, \tag{39}$$

with $F^*(0) = (\beta/H_i)^{1/2}a_i^{-y}$ and, for $v_x = 0$, $dF^*/ds(0) = (1-y)(H_i\beta)^{1/2}a_i^y$. For $y = 0$ this is Eq. (16); we verified Eq. (39) numerically for $y = 3/4$ and $y = 1$. The choice $y = 1/(2Ht)$ makes Ha^{2y} constant: in the matter era $y = 3/4$, the damping is the constant $H_0/2$ and $s = a_i^{-3/2}t_i\ln(t/t_i)$; in the radiation era $y = 1$, the damping vanishes and $s = a_i^{-2}t_i\ln(t/t_i)$.

6.2 Matter era

In the matter era with $a_i = 1$, Eq. (39) reads $d^2F^*/ds^2 = (H_0^2/16)F^* - \gamma^{n+2}(H_i/\beta)^{1+n/2}F^{*n-1}a^{3(n+2)/4} + F^{*-3}$. When the interaction dominates, the solution is approximately a parabola: freezing the interaction term at its initial value and neglecting the other two, $F^*(s)$ vanishes at

$$H_0 s_{ff}^a = \frac{\lambda}{4}\left(1 + \sqrt{1 + \frac{32}{\lambda}}\right), \tag{40}$$

where

$$\lambda = \frac{\beta^2}{\gamma^{2+n}} = \left(\frac{H_i a_i r_i}{\tilde v}\right)^2 = \frac{H_i^2(2a_i r_i)^{2-n}}{-nG_nM} = \frac{8}{\pi^2}\left(H_i t_{ff}^{ex}\right)^2 \tag{41}$$

(the last form for gravity) measures the expansion against the interaction. It defines the scale of a system: expansion dominates on large scales, $\lambda > 1$, the interaction on small scales, $\lambda < 1$. For gravity $\lambda = 2/\Delta$ (Eq. 18), so that gravitationally bound systems have $\lambda \leqslant 1$, i.e. $\Delta \geqslant 2$. Equation (40) gives $H_0 s^a_{ff} \approx \sqrt{2\lambda} = (4/\pi) H_i t^{ex}_{ff}$ for small λ and $H_0 s^a_{ff} \approx \lambda/2$ for large λ.

For gravity the factor a^{n+1} of Eq. (24) is unity, and the free fall from the Hubble flow, $d^2r/ds^2 + (H_0/2)dr/ds = -GM/[8r^2]$ with $dr/ds = 0$ at $s = 0$, depends on λ alone; we integrate it exactly. The correction factor $\lambda_c = s_{ff}/s^a_{ff}$ (Fig. 9b) tends to $\pi/4$ for $\lambda \to 0$, where the expansion is negligible and $s_{ff} = t^{ex}_{ff} a_i^{-3/2}$, and to $1/3$ for $\lambda \to \infty$, where the motion is overdamped, $dr/ds \approx -GM/(4H_0 r^2)$, and $H_0 s_{ff} = \lambda/6$. The approach to $1/3$ is slow: $\lambda_c = 0.583$, 0.413 and 0.335 for $\lambda = 9$, 10^2 and 10^4. In physical time, $H_0 s = Ht \ln(t/t_i)$ for $a_i = 1$, so that

$$\ln\left(1 + \frac{t_{ff}}{t_i}\right) = \frac{3\lambda_c}{8}\lambda\left(1 + \sqrt{1 + \frac{32}{\lambda}}\right). \tag{42}$$

Since λ depends on t_i and t^{ex}_{ff} only, the free-fall time is a function of t^{ex}_{ff}/t_i: $t_{ff}/t^{ex}_{ff} = 1.83$, 9.6 and 95 for $t^{ex}_{ff}/t_i = 1$, 3 and 5. Two regimes follow:

$$\begin{aligned} &t_{ff} \to t^{ex}_{ff} \quad (t_i \to \infty), \\ &\ln\frac{t_{ff}}{t_i} = \frac{\lambda}{4} + 2\ln\lambda + c + o(1), \quad \frac{\lambda}{4} = \frac{8}{9\pi^2}\left(\frac{t^{ex}_{ff}}{t_i}\right)^2 \quad (t_i \to 0), \end{aligned} \tag{43}$$

where the constant $c = -1.00$ was found numerically for λ between 10^5 and 10^6; the leading term $t_{ff} \approx t_i \exp[8(t^{ex}_{ff}/t_i)^2/9\pi^2]$ alone underestimates the time by the factor $\approx 0.37\lambda^2$.

6.3 Radiation era

In the radiation era ($Ht = 1/2$, $y = 1$, $Ha^2 = H^*$) Eq. (38) has no damping, and for gravity the interaction grows as $a = a_i e^{H^* s}$. Freezing F^* at its initial value in the interaction term gives

$$\begin{aligned} &F^*(s) = \frac{1}{\lambda}\left(\frac{\beta}{H^*}\right)^{1/2}\left(\lambda + 1 + H^* s - e^{H^* s}\right), \\ &\lambda = e^{H^* s^a_{ff}} - H^* s^a_{ff} - 1. \end{aligned} \tag{44}$$

The exact correction factor tends to $\pi/4$ for small λ and to 1 for large λ, but only logarithmically: $\lambda_c = 0.88$, 0.93 and 0.97 for $\lambda = 10^2$, 10^4 and 10^8. For large λ the exact equation, written in $w = H^* s - \ln\lambda$, becomes $d^2r/dw^2 = -e^w/r^2$ with $r \to r_i$ for $w \to -\infty$; it has a universal collapse point $w^* = -0.6193$. Hence $H^* s_{ff} = \ln\lambda + w^*$, and with $\lambda = (2/\pi^2)(t^{ex}_{ff}/t_i)^2$,

$$t_{ff} \approx e^{2w^*}\lambda^2 t_i = \frac{4e^{2w^*}}{\pi^4} t_i \left(\frac{t^{ex}_{ff}}{t_i}\right)^4 = 0.0119\, t_i \left(\frac{t^{ex}_{ff}}{t_i}\right)^4 \qquad (t_i \to 0), \tag{45}$$

a factor 3.45 below the value obtained with $\lambda_c = 1$.

Figure 10 shows the free-fall time of three systems with $t^{ex}_{ff} =$ 5.54 Gyr, 54.9 Myr and 8.1 Myr. A system that starts to collapse when $t_i \gg t^{ex}_{ff}$ falls in t^{ex}_{ff}; one that starts at $t_i \lesssim t^{ex}_{ff}$ takes much longer, because the Hubble flow resists the collapse.

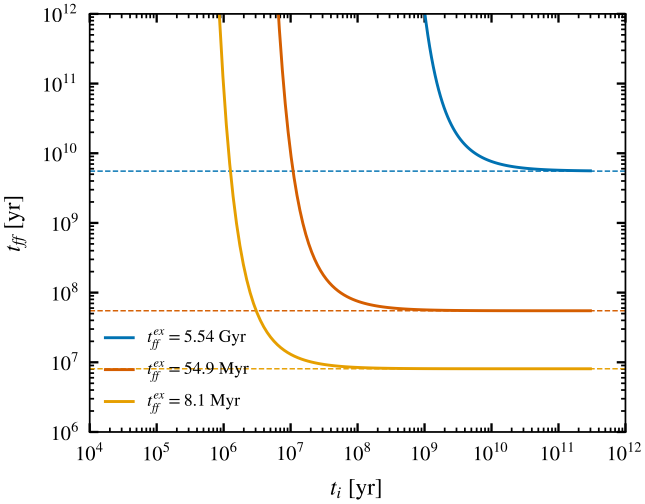


Figure 10. Free-fall time t_{ff} of a pair as a function of the time t_i at which the fall starts (matter era, exact), for static free-fall times $t^{ex}_{ff} = 5.54$ Gyr, 54.9 Myr and 8.1 Myr (dashed). A system of given size and mass falls much more slowly if it starts early.

7 TWO-PARTICLE HALOES IN AN *N*-BODY SIMULATION

7.1 Data and mapping

The smallest haloes of a simulation are pairs of particles, the closest realization of the TBCM. We use the Virgo SCDM simulation [31, 32]: 256^3 particles of mass $2.27 \times 10^{11} M_\odot/h$ in a box of 239.5 Mpc/h, $\Omega_m = 1$, $h = 0.5$, softening length 36 kpc/h. Haloes were identified with a friends-of-friends algorithm with a linking length of 0.2 Mpc/h (comoving, 0.21 of the mean interparticle separation). For each halo the catalogue [33] lists the number of particles, the specific potential and kinetic energies, the virial quantity G and the angular momentum $\mathbf{H}$. At $z = 0$ there are 956,476 two-particle haloes (and 326,734 at $z = 3$).

For an isolated pair, Eqs. (22) and (23) can be inverted. For gravity,

$$\begin{aligned} &F_a = -\frac{P_y}{2K_p}\left[1 + \left(\frac{G}{|\mathbf{H}|}\right)^2\right], \qquad x = -\frac{P_y F_a t}{|\mathbf{H}|}, \\ &F'_a = \frac{G}{|\mathbf{H}|F_a} + \frac{Ht}{x}F_a, \qquad \Delta = \frac{2x^2}{F_a^3 (Ht)^2}, \\ &C_1 = -\frac{(F'_a F_a)^2 + 1 - 2F_a}{F_a^2}, \end{aligned} \tag{46}$$

which give the state (x, F_a, F'_a) of each pair in the units of Eq. (19), its current density relative to the background and the constant of its limiting Kepler orbit (Eq. 30); $C_1 > 0$ means that the pair is bound, including the Hubble flow. Equations (46) recover the state of exact pair solutions to 2×10^{-9}. In the simulation the energies and momenta are in physical units (P_y from physical distances, K_p, G and $\mathbf{H}$ from peculiar velocities, t the age at the redshift of the snapshot). As an independent check, the Δ of Eq. (46) equals the overdensity computed from the separation of the two particles alone to 1×10^{-3} at every redshift.

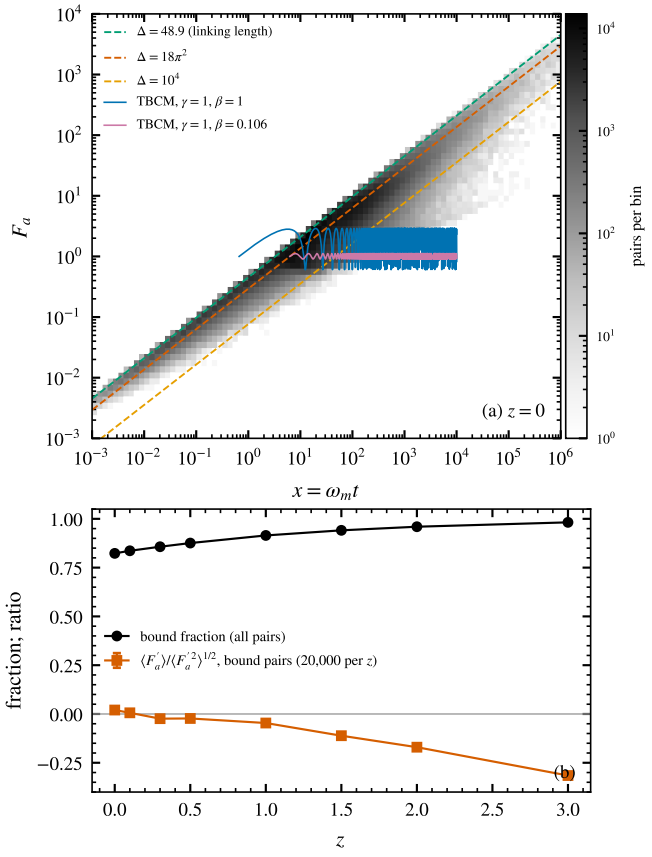


Figure 11. Two-particle haloes of the simulation. (a) Number of pairs in the (x, F_a) plane at $z = 0$ (Eq. 46), with lines of constant overdensity (dashed; $\Delta = 48.9$ corresponds to the linking length) and two TBCM solutions. (b) Fraction of bound pairs ($C_1 > 0$) and the mean physical radial velocity of the bound pairs, $\langle F'_a\rangle/\langle F'^2_a\rangle^{1/2}$ (20,000 random pairs per redshift), as functions of z.

Table 1. Two-particle haloes of the simulation: number, percentage bound ($C_1 > 0$), median overdensity, median C_1 of the bound pairs and their mean radial velocity $\langle F'_a\rangle/\langle F'^2_a\rangle^{1/2}$ (statistical error ≈0.008).

z	N	bound (%)	med. Δ	med. C_1	$\langle F'_a\rangle$
0	956,476	82.3	222	0.42	+0.02
0.1	974,227	83.6	206	0.42	+0.01
0.3	996,820	85.7	187	0.41	-0.02
0.5	999,254	87.6	173	0.42	-0.02
1	945,770	91.5	160	0.42	-0.05
1.5	808,601	94.1	146	0.39	-0.11
2	651,550	96.0	168	0.37	-0.17
3	326,734	98.2	184	0.30	-0.32

7.2 Results

Figure 11a shows the pairs at $z = 0$ in the (x, F_a) plane. Their lower boundary is the line $\Delta = 48.9$ set by the linking length: the halo finder selects pairs with a separation below 0.2 Mpc/h, so the distribution of Δ (median 222) reflects the halo finder and is not a test of the model. The dynamical state, however, is. 82 per cent of the pairs are bound at $z = 0$, and the fraction increases with redshift to 98 per cent at $z = 3$ (Table 1). The median C_1 of the bound pairs is 0.42, i.e. eccentric orbits ($e \approx 0.76$).

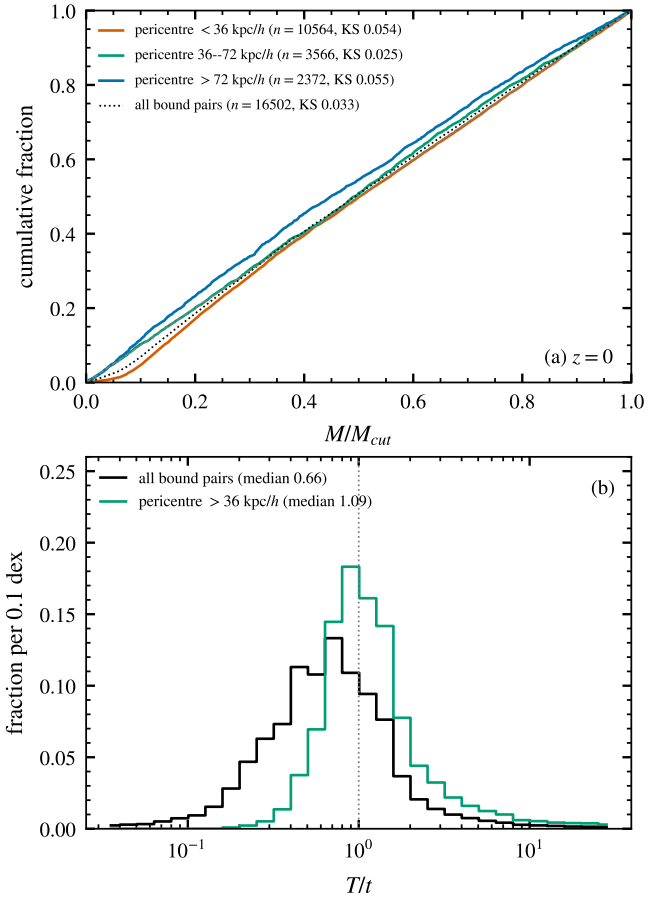


Figure 12. (a) Cumulative distribution of the orbital phase M/M_{cut} of the bound two-particle haloes at $z = 0$ (Eq. 47), for pericentres inside the softening length, between one and two softening lengths, and beyond, and for all bound pairs; phase-mixed pairs follow the diagonal. KS: maximum deviation from the diagonal. (b) Distribution of the orbital period T of the bound pairs in units of the cosmic time t.

Stable clustering of pairs. If bound pairs have reached a fixed physical size, their physical radial velocity F'_a averages to zero. At $z = 0$ the mean is +0.02 of its rms (Fig. 11b): the bound pairs are stable. At higher redshift the mean becomes negative, −0.05 at $z = 1$, −0.17 at $z = 2$ and −0.32 at $z = 3$: the pairs are still falling in.

Phase mixing. If the bound pairs move on their limiting Kepler orbits with random phases, the time since the last pericentre passage, measured by the mean anomaly $M = \eta - e\sin\eta$ with $1 - C_1 F_a = e\cos\eta$, is uniformly distributed. The halo finder counts a pair only while its separation is below the linking length, i.e. for $M \leqslant M_{cut}$, where M_{cut} corresponds to $F_a \leqslant F_a \times 0.1\,\mathrm{Mpc}\,h^{-1}/R$ (R the half-separation). Phase-mixed pairs therefore have

$$M/M_{cut} \text{ uniform on } [0, 1]. \tag{47}$$

For the 16,502 bound pairs of a random sample of 20,000 at $z = 0$ the maximum deviation of the cumulative distribution from uniform is 0.033 (Fig. 12a), larger than the 95 per cent Kolmogorov–Smirnov limit 0.011. The deviation is a deficit near pericentre, and it comes from the 64 per cent of the pairs whose pericentre lies inside the softening length (deviation 0.054; 0.047 of them in the first tenth of M_{cut} instead of 0.1), where the force is not Keplerian. Pairs with a pericentre between one and two softening lengths (22 per cent) are

close to phase-mixed (deviation 0.025, limit 0.023); pairs with wider orbits (14 per cent; 0.055, limit 0.028) have periods comparable to the age and are not yet mixed.

Orbital period. The orbital period of a bound pair is $T = 2\pi/(C_1^{3/2}\omega_m)$, i.e. $T/t = 2\pi/(C_1^{3/2}x)$. The median is 0.66 for all bound pairs and 1.09 (16th–84th percentiles 0.68–2.19) for the pairs whose pericentre is resolved (Fig. 12b): the orbital period of the bound pairs is close to the cosmic time, the condition behind $\Delta_c = 18\pi^2$ (Section 5.5). The linking length favours small orbits, so the distribution is not that of all bound pairs of particles.

8 CONNECTIONS WITH EXISTING THEORIES

8.1 Stable clustering and its generalization

For a bound pair the pairwise peculiar velocity along the separation $2\mathbf{R}$ is

$$\Delta u_L = -2HR + 2\dot{R}, \tag{48}$$

where $\dot{R} \propto F_a'$ oscillates about zero. For an ensemble of bound pairs with random phases $\langle\dot{R}\rangle = 0$, so $\langle\Delta u_L\rangle = -Hr_{\rm sep}$ with $r_{\rm sep} = 2R$: the stable clustering hypothesis follows from the type II solutions for every long-range exponent, and Section 7 shows that the bound two-particle haloes of the simulation satisfy it at $z = 0$. In the pairwise statistics of all particles, $-\langle\Delta u_L\rangle/(Har) = 1.03$ over $0.05 < r < 0.2$ Mpc/h at $z = 0$ (r comoving), but 1.35 over $0.2 < r < 0.5$ Mpc/h, where pairs are still falling in.

Write $\Delta u_L = A + B$, with the infall part A, $\langle A\rangle = -Har$, and a symmetric random part B. To first order in r,

$$\langle\Delta u_L^{2m}\rangle = \langle B^{2m}\rangle, \qquad \langle\Delta u_L^{2m+1}\rangle = (2m+1)\langle A\,B^{2m}\rangle. \tag{49}$$

If A is independent of B, as for bound pairs with random phases, this gives the generalized stable clustering hypothesis (GSCH)

$$R_m \equiv \frac{\langle\Delta u_L^{2m+1}\rangle}{\langle\Delta u_L^{2m}\rangle\langle\Delta u_L\rangle} = 2m+1, \tag{50}$$

which reduces to the SCH for $m = 0$. In general $R_m/(2m+1) = \langle AB^{2m}\rangle/(\langle A\rangle\langle B^{2m}\rangle)$, the infall velocity weighted by B^{2m} relative to its mean; it exceeds unity and grows with m if pairs with a large random velocity fall in faster.

We test Eq. (50) with the moments $\langle\Delta u_L^k\rangle$, $k = 1$–9, of all particle pairs of the same simulation at separations 0.001–119.75 Mpc/h [34, 35]. Below the softening length the ratios are noisy (Fig. 13a). Between 0.04 and 0.2 Mpc/h, where stable clustering holds, R_1 is close to the GSCH value 3, while R_2, R_3 and R_4 lie far above 5, 7 and 9: the medians over $0.05 < r < 0.2$ Mpc/h are 3.3, 6.6, 15.4, 31.2 for $m = 1$–4, and over $0.2 < r < 0.5$ Mpc/h 3.4, 7.0, 13.1, 21.7 (Fig. 13b). The GSCH thus holds for $m = 1$ to about 10 per cent and fails for $m \geqslant 2$: pairs with a large random velocity fall in faster than the mean. At higher redshift the ratios over 0.2–0.5 Mpc/h are closer to $2m+1$ (at $z = 1$: 2.9, 4.8, 7.0, 8.9), but there the mean infall exceeds the Hubble flow by a factor 2.1, outside the regime assumed in the derivation.

The second moment at small separation fixes the random part. Writing $\langle\Delta u_L^2\rangle = \alpha_u u_0^2 a^{\beta_u}$, the simulation gives $\alpha_u = 3.6$ and $\beta_u = 1.31$ at $r = 0.1$ Mpc/h for $z \leqslant 2$ ($\langle\Delta u_L^2\rangle/u^2 = 3.56$ and 3.50 at $z = 0$ and 1, with u the one-dimensional rms velocity of all particles), instead of $\alpha_u = 2$ (uncorrelated pair velocities) and $\beta_u = 3/2$ ($u^2 \propto t$, a steady energy cascade). With the generalized kurtosis $K_{2m} = \langle\Delta u_L^{2m}\rangle/\langle\Delta u_L^2\rangle^m$, the GSCH would fix all odd moments at small r as $\langle\Delta u_L^{2m+1}\rangle = -(2m+1)K_{2m}\alpha_u^m u_0^{2m} a^{m\beta_u} Har$; the simulation shows that this holds for $m = 1$ only.

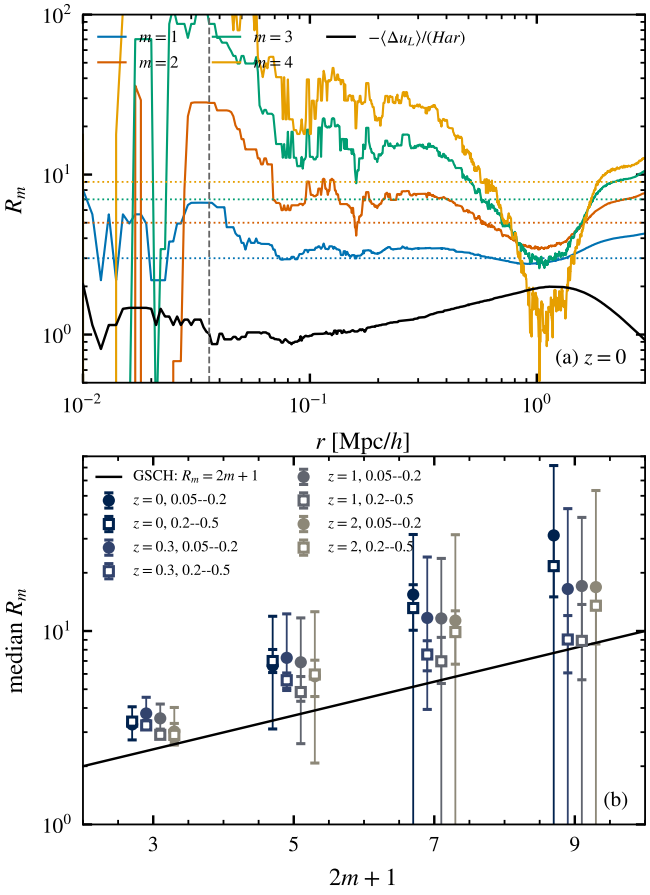


Figure 13. Generalized stable clustering in the N-body simulation. (a) $R_m(r)$ (Eq. 50) at $z = 0$ for $m = 1$–4 (moving median over 21 bins of 1 kpc/h; dotted: $2m+1$) and the stable-clustering ratio $-\langle\Delta u_L\rangle/(Har)$; dashed vertical line: softening length. (b) Medians of R_m over 0.05–0.2 (filled) and 0.2–0.5 Mpc/h (open) at $z = 0$, 0.3, 1 and 2, with the 16th–84th percentiles over r; solid line: GSCH.

8.2 Violent relaxation

The relative motion depends on the masses only through M, and the peculiar velocities of the two masses are $\mu a\dot{\mathbf{r}}$ and $-(2-\mu)a\dot{\mathbf{r}}$ at all times. The ratio of their kinetic energies therefore stays m_2/m_1, and the ratio of their potential energies stays 1: no energy is exchanged between the masses, and the specific energy is the same for both. A two-body collapse does not segregate masses, as in violent relaxation [36] and unlike collisional two-body relaxation, which drives energy equipartition and lets massive particles sink [37, 38].

8.3 Spherical collapse model

The SCM follows a spherical shell of radius R around an overdensity of mass M, $\ddot{R} = -GM/R^2$, starting with the Hubble flow, $\dot{R} = HR$; its solution is the cycloid $R = A(1-\cos\vartheta)$, $t = B(\vartheta - \sin\vartheta)$ with $A^3 = GMB^2$. In the variables of Eq. (19) it corresponds to keeping only the interaction term, $F_a'' = -F_a^{-2}$, with $F_a(0) = 0$: the SCM neglects both the background force (term 1) and the centrifugal force (term 3). Equation (10) keeps the homogeneous background, whose force $(\ddot{a}/a)R$ acts on the pair, whereas in the SCM the shell feels the total enclosed mass in an otherwise empty space. The TBCM can therefore be viewed as a nonradial collapse model with an exactly

conserved angular momentum. The SCM has no bound oscillating state (no stable clustering) and reaches virial equilibrium only by assumption at the critical density; the TBCM contains the bound state, the energy cycle and the free fall in one equation. Both lead to $\Delta_c = 18\pi^2$ through the orbital-period identity of Section 5.5.

8.4 Scope

The TBCM treats each merger in isolation; tidal fields, dynamical friction and the internal structure of the merging haloes are not included. Estimates of the energy of large haloes that grow by mergers, and links to the isothermal density profile, would need the further assumption that each merger adds the kinetic energy of a virialized pair; they are not a result of the two-body dynamics and are not used here.

9 CONCLUSIONS

(i) The two-body problem with a power-law potential in a power-law expanding background reduces exactly to Eq. (19) for the physical size F_a of the pair, with the dimensionless parameters n, γ, β and v_x/v_y; energies and momenta follow from F_a.

(ii) Three types of solution exist: expansion with the background (type I) and fall to the centre (type III) for short-range forces, and bound pairs oscillating about a fixed physical size (type II) for long-range forces. Stable clustering is a property of every long-range force. The phase diagrams agree with the author's independent maps.

(iii) For gravity the late-time state is a Kepler orbit, solved in closed form; the energy of the pair oscillates about the Kepler value with phases of durations $T_{1,2} = (\pi \mp 2e)/C_1^{3/2}$ and power-law exponents $\pm 4Ht\,e/(\pi \mp 2e)$. Pairs that start in virial balance with the Hubble flow always turn around, at a time that grows steeply with β.

(iv) The critical $\beta_c = \sqrt{2+n}\,Ht/(2\pi)$ exists for every long-range exponent, and $\Delta_c = 18\pi^2$ ($32\pi^2$ in the radiation era) is equivalent to an orbital period equal to the cosmic time, the same condition as in the SCM.

(v) The free-fall time is a function of t_{ff}^{ex}/t_i alone. The correction factor of the constant-force approximation tends to $\pi/4$ on small scales and to $1/3$ (matter era) or, logarithmically, to 1 (radiation era) on large scales; we give the exact asymptotes.

(vi) In the N-body simulation, most two-particle haloes are bound pairs; at $z = 0$ their mean radial velocity vanishes, their orbital phases are close to mixed where the orbits are resolved, and their orbital periods are close to the cosmic time. At $z \geqslant 2$ the pairs are still falling in.

(vii) The generalized stable clustering hypothesis holds for the third moment of the pairwise velocity but not for higher moments, which shows that infall and random pairwise velocities are correlated on small scales.

DATA AVAILABILITY

The halo catalogues and the pairwise velocity statistics of the simulation are part of the dark matter flow datasets on Zenodo [33, 34], with the accompanying presentation slides [39] and all data files [40]. The reproducible package of this paper contains the manuscript source, the Python solver and pipeline for every figure and number, the reduced two-particle-halo data, all results as CSV files, the author's original MATLAB scripts, an equation audit and a verification log.

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